

Fair Committee Selection with Approval Voting and Marginal Quotas

Abstract

We study a fair committee selection problem under approval voting. Each voter submits an approval ballot over candidates, and each candidate has two binary attributes. We impose marginal quotas on each attribute value separately, so a candidate can satisfy more than one quota at the same time. Our goal is to select a committee of fixed size k with maximum approval support while meeting all marginal fairness constraints. Because the attributes are binary, the problem can be solved exactly by enumerating all feasible type-count vectors.

1 Introduction

Approval voting is a natural model for committee selection: each voter approves any number of candidates, and the committee is chosen to reflect these approvals. In many applications, candidates also have protected attributes such as gender and race, and one wants the final committee to be fair with respect to these attributes.

In this paper, we consider a simple and practical version of fair committee selection. Each candidate has two binary attributes, and fairness is expressed through *marginal quotas*. That means we impose lower bounds on each attribute value separately. A selected candidate contributes to every quota corresponding to its attribute values. For example, if a candidate is black and male, then that candidate counts toward both the black quota and the male quota.

The advantage of marginal quotas is that they are easy to interpret and easy to compute with. Since the number of attribute types is constant, we can give an exact polynomial-time algorithm based on enumerating all feasible type-count vectors.

2 Problem

We are given:

- a set of candidates C ,
- a set of voters V ,
- a target committee size k ,
- for each voter $v \in V$, an approval ballot $A_v \subseteq C$,
- for each candidate $c \in C$, two binary attributes $X(c), Y(c) \in \{0, 1\}$.

The goal is to choose a committee $W \subseteq C$ with $|W| = k$ that maximizes approval support and satisfies marginal quotas on both attributes.

For the first attribute, we require

$$|W \cap C_{X=0}| \geq q_{X=0}, \quad |W \cap C_{X=1}| \geq q_{X=1},$$

and for the second attribute,

$$|W \cap C_{Y=0}| \geq q_{Y=0}, \quad |W \cap C_{Y=1}| \geq q_{Y=1}.$$

The approval score of a candidate is

$$\text{score}(c) = |\{v \in V : c \in A_v\}|.$$

The approval score of a committee is

$$\text{score}(W) = \sum_{c \in W} \text{score}(c).$$

3 Setup

Because the two attributes are binary, every candidate belongs to exactly one of four types:

$$C_{00}, C_{01}, C_{10}, C_{11},$$

where

$$C_{ab} = \{c \in C : X(c) = a, Y(c) = b\}.$$

A committee W is feasible if:

$$\begin{aligned} |W| &= k, \\ |W \cap C_{X=0}| &\geq q_{X=0}, \quad |W \cap C_{X=1}| \geq q_{X=1}, \\ |W \cap C_{Y=0}| &\geq q_{Y=0}, \quad |W \cap C_{Y=1}| \geq q_{Y=1}. \end{aligned}$$

A useful way to represent a committee is by the count vector

$$x = (x_{00}, x_{01}, x_{10}, x_{11}),$$

where x_{ab} is the number of selected candidates from C_{ab} . Then feasibility means:

$$\begin{aligned} x_{00} + x_{01} + x_{10} + x_{11} &= k, \\ x_{00} + x_{01} &\geq q_{X=0}, \quad x_{10} + x_{11} \geq q_{X=1}, \\ x_{00} + x_{10} &\geq q_{Y=0}, \quad x_{01} + x_{11} \geq q_{Y=1}. \end{aligned}$$

For any fixed feasible vector x , the best committee realizing that vector is obtained by selecting the top-scoring candidates inside each type.

4 Algorithm

Fair Committee Selection with Marginal Quotas

Require: Candidates C , voters V , committee size k , quotas $q_{X=0}, q_{X=1}, q_{Y=0}, q_{Y=1}$

Ensure: A feasible committee W of size k with maximum approval score

- 1: Compute $\text{score}(c)$ for every candidate $c \in C$
- 2: Partition candidates into $C_{00}, C_{01}, C_{10}, C_{11}$
- 3: Sort each group in non-increasing order of $\text{score}(c)$
- 4: $\text{bestScore} \leftarrow -\infty$
- 5: $\text{bestCommittee} \leftarrow \emptyset$
- 6: **for** $x_{00} = 0$ to $|C_{00}|$ **do**
- 7: **for** $x_{01} = 0$ to $|C_{01}|$ **do**
- 8: **for** $x_{10} = 0$ to $|C_{10}|$ **do**

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9:       $x_{11} \leftarrow k - x_{00} - x_{01} - x_{10}$ 
10:    if  $x_{11} < 0$  or  $x_{11} > |C_{11}|$  then
11:      continue
12:    end if
13:    if  $x_{00} + x_{01} < q_{X=0}$  then
14:      continue
15:    end if
16:    if  $x_{10} + x_{11} < q_{X=1}$  then
17:      continue
18:    end if
19:    if  $x_{00} + x_{10} < q_{Y=0}$  then
20:      continue
21:    end if
22:    if  $x_{01} + x_{11} < q_{Y=1}$  then
23:      continue
24:    end if
25:     $W \leftarrow$  top  $x_{00}$  candidates from  $C_{00}$ 
26:     $\cup$  top  $x_{01}$  candidates from  $C_{01}$ 
27:     $\cup$  top  $x_{10}$  candidates from  $C_{10}$ 
28:     $\cup$  top  $x_{11}$  candidates from  $C_{11}$ 
29:     $total \leftarrow \sum_{c \in W} \text{score}(c)$ 
30:    if  $total > bestScore$  then
31:       $bestScore \leftarrow total$ 
32:       $bestCommittee \leftarrow W$ 
33:    end if
34:  end for
35: end for
36: end for
37: return  $bestCommittee$ 

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5 Example

Suppose $k = 3$ and the quotas are:

$$q_{X=0} = 1, \quad q_{X=1} = 1, \quad q_{Y=0} = 1, \quad q_{Y=1} = 1.$$

Consider four candidates:

$$c_1 \in C_{00}, \quad c_2 \in C_{01}, \quad c_3 \in C_{10}, \quad c_4 \in C_{11}.$$

If we select:

$$W = \{c_1, c_2, c_4\},$$

then:

$$\begin{aligned}
|W \cap C_{X=0}| &= 2, & |W \cap C_{X=1}| &= 1, \\
|W \cap C_{Y=0}| &= 1, & |W \cap C_{Y=1}| &= 2.
\end{aligned}$$

So this committee satisfies all marginal quotas. Notice that c_4 , which is in C_{11} , contributes to both the $X = 1$ and $Y = 1$ quotas.

6 Time Complexity

Let $n = |C|$ and $m = |V|$.

- Computing approval scores takes $O(nm)$ time.
- Partitioning and sorting the four groups takes $O(n \log n)$ time.
- The enumeration step checks at most $O(k^3)$ feasible count vectors.

Therefore the total running time is

$$O(nm + n \log n + k^3).$$

So the algorithm is polynomial time and exact.

7 Hardness of the Generalized p -Attribute Problem

We now consider the generalized model where the number of binary attributes p is part of the input. In this setting, the problem becomes NP-hard.

The fair committee selection problem with approval voting and marginal quotas over p binary attributes is NP-hard when p is part of the input.

Proof. We reduce from the SET COVER problem. Let $(U, \mathcal{S}, k)$ be an instance of SET COVER, where $U = \{e_1, \dots, e_p\}$ is the universe of elements and $\mathcal{S} = \{S_1, \dots, S_n\}$ is a collection of subsets of U .

We construct an instance of fair committee selection as follows: - For each set $S_i \in \mathcal{S}$, create a candidate c_i . - Introduce p binary attributes, one for each element in U . - For each candidate c_i and attribute j , define $X_j(c_i) = 1$ if $e_j \in S_i$, and 0 otherwise. - Set the committee size to k . - For each attribute j , impose a quota requiring at least one selected candidate with $X_j = 1$, i.e., $q_{X_j=1} = 1$.

We assign equal approval scores to all candidates, so the objective is irrelevant and the problem reduces to feasibility.

Then, a feasible committee of size k exists if and only if there is a set cover of size at most k . This completes the reduction. $\square$

8 Interpretation

This model uses *marginal quotas*, not intersectional quotas. That means a candidate contributes to every attribute-value quota that matches its own attributes. So a black male candidate counts toward both the black quota and the male quota.

This makes the model simple, credible, and easy to explain in a meeting. It also gives a clean exact algorithm because the number of candidate types is fixed.